Boiling and melting points

☞ **1** Read the following passage and circle the correctly spelt word in each pair.

The (boiling/boling) point of a substance is the (temperatur/temperature) at which a substance changes state from a (liquid/liqid) to a gas. The melting point of a substance is the temperature at which a substance changes state from a (soilid/solid) to a liquid. Different substances have different melting and boiling points. For (pure/prue) water, the melting point is 0°C and the boiling point is 100°C.

If the temperature is above a substance's boiling point, the substance is a (gas/gase). If the temperature is between a substance's melting and boiling point, the substance is a liquid. Below its melting point, a substance is a solid. Substances that have a (mealting/melting) and boiling point below room temperature (around 20°C) are usually gases.

2 Use the information in the following table to answer the questions below.

Substance	Melting point °C	Boiling point °C
Iron	1535	2750
Mercury	–39	357
Pure water	0	100
Oxygen	–218	–183

a What state of matter is oxygen at 25°C? _____

b At 25°C, what state of matter is mercury? _____

c Is 25°C above or below the melting point of iron? _____

d What temperature must iron be heated to before it will change from a gas to a liquid? _____

e What temperature does oxygen have to be before it will change from a gas to a liquid? _____

f Is 25°C above the melting point but below the b_____ p_____ for water? _____

g Which substance listed in the table has the lowest melting point? _____

© Janine Haeusler and Blake Education 2014
© Blake Education ISBN 978 1 92571 061 8 Achieve! Science Science understanding - Chemistry Years 7-8